OPTIMAL CONTRACT AND GENERAL CARBON THEORY FOR FLEXIBLE COMPUTING LOADS: EXISTENCE AND UNIQUENESS, A UNIFORM HOMOTOPY CONDITION, THE GENERAL-F EMISSION KEYSTONE, CONNECTEDNESS, AND SHARPNESS

TEAM A1 — ROLE 1 (OPTIMAL CONTRACT AND GENERAL CARBON THEORY)

ABSTRACT. This is the Role 1 Week 2 deliverable. It hardens the Week 1 carbon theory (`role1_carbon_theory.tex`) into five results—four proved, one resolved numerically—taking the same model and notation and staying in Role 1's large-FCL mandate (the one small-FCL touch is flagged as a conjecture pending Role 4). *One*, the regulated program has a maximizer, it is unique, and the within-period dispatch $d^*(x) = \min\{(K_F - x)_+, K_M\}$ is the unique optimal rule; uniqueness comes not from global concavity (the profit is submodular in (K_R, K_F), so it fails) but from a *recursive-triangular* structure—the distribution-free cutoffs pin K_F and then K_M as unique functions of K_R, after which a single scalar diagonal-dominance (reduced second-order) condition pins K_R. *Two*, support-wide monotonicity of the shift ratio $g = F_{K_R}/f$ makes the realized two-point endpoint inequality hold *uniformly* along the whole C_M-homotopy, even as the dispatch window moves—the exact content the Week 1 "integrate-downward" step needs. *Three* (the keystone), substituting the equilibrium map $K^*(C_R)$ into the exact inequality (C) yields the endogenous carbon condition, and we give a *primitive* sufficient condition—the endpoint shift-ratio condition plus a two-zone density (tail/skew) bound—that would make carbon fall at a *finite interior* optimum. The point is what this checkable inequality then reveals: it is essentially *never activated*. We prove an irreducible fossil floor (a program with $K_M^* > K_R^*$ draws at least $K_M^* - K_R^*$ fossil, since the FCL exceeds the maximal renewable surplus $\le K_R^*$ and cannot be fully surplus-powered), and an exhaustive numerical search over the Beta scale-mixture family then finds $\Delta \le 0$ at *every* large-FCL interior optimum located ($\Delta^{\max} \approx -1.4$); we conjecture, but do not analytically prove, that this holds family-wide, and that the small-FCL case (Role 4's regime) also fails because there the no-FCL grid is already near fossil-free. Carbon strictly falls only in the cheap-renewable corner $K_R^* \to \infty$. So the honest reading is a *negative* one: interior carbon reduction is rare and unwitnessed; it is a corner phenomenon. *Four*, the carbon-reduction set $\{C_R : \Delta(C_R) \ge 0\}$ is bounded above and non-empty—its non-emptiness coming from that power-law corner, not from an interior optimum—and is a single interval under an explicit monotone-renewable-shadow-value condition that, we note, already encodes the single-crossing property it certifies. *Five*, the sharp necessary-and-sufficient condition for $K_R^* > K_R^\circ$ is the *path integral* of the realized endpoint gap; pointwise-monotone g is sufficient but not necessary; and the two-sided bounds on the carbon saving are tight, attained by explicit degenerate conditional laws. Every identity is checked against quadrature of the profit integral in `verify_week2.py`.

1. SETUP

We keep the model and notation of `role1_carbon_theory.tex`. A regulated utility chooses renewable capacity K_R, dispatchable fossil K_F, and flexible-computing-load (FCL) capacity K_M. Residual demand is $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density $f(x; K_R) > 0$ on $(-K_R, \bar{D})$, and $F_{K_R} := \partial F/\partial K_R \ge 0$, extended by $F \equiv 0$ below $-K_R$ and $F \equiv 1$ above $\bar{D}$. Per-MWh: FCL revenue R, fossil cost c, price p, blackout penalty β; amortized capital costs C_R, C_F, C_M.

Assumption 1 (Regularity, as in Week 1)**.** $R > c + C_M$, $\quad R < c + C_F + C_M$, $\quad \beta + p > c + C_F$, $\beta + p > R$.

Date: Working results, Week 2.

Under the dispatch rule $d(x) = \min\{(K_F - x)_+, K_M\}$ the unified profit is

$$\Pi = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F-K_M}^{K_F} F\,dx - c\int_{-K_M}^{K_F-K_M}(1-F)\,dx - (\beta+p)\int_{K_F}^{\bar{D}}(1-F)\,dx - C_F K_F - C_R K_R - C_M K_M, \quad (1)$$

with gradient (Week 1, Lemma 1)

$$g_M := \Pi_{K_M} = (R-c)F(K_F - K_M) + cF(-K_M) - C_M, \quad (2)$$

$$g_F := \Pi_{K_F} = R[F(K_F) - F(K_F - K_M)] - c[1 - F(K_F - K_M)] + (\beta+p)[1 - F(K_F)] - C_F, \quad (3)$$

$$g_R := \Pi_{K_R} = R\int_{K_F-K_M}^{K_F} F_{K_R}\,dx + c\int_{-K_M}^{K_F-K_M} F_{K_R}\,dx + (\beta+p)\int_{K_F}^{\bar{D}} F_{K_R}\,dx - C_R. \quad (4)$$

We work in the *large-FCL interior* regime $K_M \geq K_R$ (so $F(-K_M) = f(-K_M) = F_{K_R}(-K_M) = 0$), the regime in which the Week-1 carbon theory lives; throughout, "interior" means $0 < K_R < \infty$ and the cutoffs below sit strictly inside $(0,1)$. Write $a := K_F - K_M$ for the lower dispatch endpoint and recall the distribution-free cutoff *values* (Week 1, Lemma 2)

$$\varphi_1 := F(a; K_R) = \frac{C_M}{R-c}, \qquad \varphi_2 := F(K_F; K_R) = \frac{\beta + p - c - C_F - C_M}{\beta + p - R}, \qquad \varphi^\circ := F(K_F^\circ; K_R^\circ) = \frac{\beta + p - c - C_I}{\beta + p - c} \quad (5)$$

Under Assumption 1, $0 < \varphi_1 < 1$ and $0 < \varphi_2 < 1$; the interior FCL (adoption) condition $K_M > 0$ is $\varphi_2 > \varphi_1$, equivalently $C_M < \bar{C}_M := (R-c)\varphi^\circ$.

2. Existence, uniqueness, and contract optimality

The optimum is not delivered by concavity. By the Week-1 cross-partial $\Pi_{K_R K_F} = (R - \beta + p)F_{K_R}(K_F) - (R-c)F_{K_R}(a) \leq 0$, Π is *submodular* in (K_R, K_F), so the Hessian is not globally negative semidefinite and global concavity is unavailable. Uniqueness instead follows from a recursive-triangular structure together with a single scalar condition.

2.1. **Dispatch optimality.**

Lemma 1 (Unique optimal dispatch via strict $R > c$)**.** *Fix capacities (K_R, K_F, K_M) and a residual-demand realization x. Let the within-period payoff from serving s units of consumer demand and dispatching d units of FCL be*

$$\pi_x(s,d) = p\,s - c\,(s+d) + R\,d - \beta\,((x)_+ - s), \qquad 0 \leq s \leq \min\{(x)_+, K_F\},\ \ 0 \leq d \leq K_M,\ \ s+d \leq K_F.$$

Under Assumption 1 the unique maximizer is $s^* = \min\{(x)_+, K_F\}$ and $d^*(x) = \min\{(K_F - x)_+, K_M\}$.

Proof. π_x is affine in (s,d) with $\partial\pi_x/\partial s = \beta + p - c > 0$ and $\partial\pi_x/\partial d = R - c > 0$, both strict under Assumption 1 ($\beta + p > c + C_F \geq c$ and $R > c + C_M > c$). A strictly increasing affine objective on a compact polytope is maximized at the unique vertex reached by pushing both coordinates up to their constraints: first $s^* = \min\{(x)_+, K_F\}$ (serving consumers dominates because $\beta + p - c > R - c$, so consumer service takes priority for fossil headroom), then d up to the remaining headroom, $d^* = \min\{K_M, K_F - s^*\} = \min\{(K_F - x)_+, K_M\}$. Strictness of both slopes makes the maximizer unique; at $R = c$ the d-slope vanishes and any $d \in [0, \min\{(K_F - x)_+, K_M\}]$ would be optimal. Hence strict $R > c$ is exactly what pins d^*.[1] □

[1] Verified in `verify_week2.py` T1a: a grid search of π_x over (s,d) returns (s^*, d^*) at every sampled x.

2.2. **Existence.**

Proposition 1 (A maximizer exists)**.** Π *is continuous on* $\mathbb{R}^3_+$*, and there is a box* $[0,\bar{K}]^3$ *that no maximizing sequence leaves. Hence the set of maximizers is nonempty and compact.*

Proof. Continuity is immediate: each integrand in (1) is bounded and $F(\cdot;\cdot)$ is jointly continuous. We do not claim coercivity along all rays; we show each *own* partial turns strictly negative once its own capacity is large. As $K_M \to \infty$ (others fixed), $K_F - K_M \to -\infty$, so $F(K_F - K_M) \to 0$ and $g_M \to -C_M < 0$. As $K_F \to \infty$, $F(K_F - K_M) \to 1$ and $g_F \to -C_F < 0$. As $K_R \to \infty$, $F_{K_R}(x; K_R) \to 0$ pointwise and, for the bounded-density families we use, $\int F_{K_R}\, dx \to 0$ by dominated convergence ($F_{K_R} = \mathbb{E}[\tilde{S} f_{\tilde{D}}]$ is dominated by $\sup f_{\tilde{D}} < \infty$), so $g_R \to -C_R < 0$. Thus there is $\bar{K} < \infty$ with $\Pi_{K_i} < 0$ whenever $K_i > \bar{K}$ (others in $[0,\bar{K}]$): on each face $K_i = \bar{K}$ the gradient points strictly *inward*, so Π has no larger value outside $[0,\bar{K}]^3$ than on its boundary, and no maximizing sequence escapes the box. By Weierstrass on the compact box a maximizer exists.[2] □

2.3. **Uniqueness by recursive triangularity.** Fix K_R. By (5) the equations $g_M = 0,\ g_F = 0$ are equivalent to $F(a; K_R) = \varphi_1$ and $F(K_F; K_R) = \varphi_2$, two *distribution-free* targets. Since $x \mapsto F(x; K_R)$ is a strictly increasing bijection from $(-K_R, \bar{D})$ onto $(0,1)$, each target has a unique preimage:

$$K_F(K_R) = F^{-1}(\varphi_2; K_R), \qquad a(K_R) = F^{-1}(\varphi_1; K_R), \qquad K_M(K_R) = K_F(K_R) - a(K_R) > 0, \tag{6}$$

all C^1 in K_R (implicit function theorem, $\partial F/\partial x = f > 0$). This is the triangle: levels one and two (K_F, K_M) are pinned by K_R with no distributional input. Substituting (6) into g_R defines the reduced renewable condition

$$\Psi(K_R) := g_R(K_R, K_F(K_R), K_M(K_R)), \qquad \Psi(K_R^*) = 0. \tag{7}$$

Lemma 2 (The (K_F, K_M) block is negative definite given interiority)**.** *At any large-*FCL *interior point (so* $f(a), f(K_F^*) > 0$*, the cutoffs strictly inside the support) the* 2×2 *Hessian block in* (K_F, K_M)*,* $\mathcal{M} = \begin{pmatrix} \Pi_{K_F K_F} & \Pi_{K_F K_M} \\ \Pi_{K_F K_M} & \Pi_{K_M K_M} \end{pmatrix}$*, satisfies* $\Pi_{K_F K_F} < 0$*,* $\Pi_{K_M K_M} < 0$*, and*

$$\det \mathcal{M} = (R-c) f(a)\, (\beta + p - R) f(K_F) \ > \ 0,$$

so $\mathcal{M} \prec 0$ *under Assumption 1 given interiority (no* distributional *condition beyond* $f > 0$ *at the cutoffs).*

Proof. From (2)–(3) in the large-FCL regime, $\Pi_{K_M K_M} = -(R-c) f(a) < 0$, $\Pi_{K_F K_M} = (R-c) f(a)$, and $\Pi_{K_F K_F} = (R - \beta + p) f(K_F) - (R-c) f(a) < 0$ (both terms negative since $R < \beta + p$, $R > c$). Then $\det \mathcal{M} = \Pi_{K_F K_F} \Pi_{K_M K_M} - \Pi^2_{K_F K_M} = [-(R-c) f(a)]\, (R - \beta + p) f(K_F) + (R-c)^2 f(a)^2 - (R-c)^2 f(a)^2 = (R-c) f(a)(\beta + p - R) f(K_F) > 0$. Negative diagonal and positive determinant give $\mathcal{M} \prec 0$.[3] □

Condition 1 (Reduced second-order / diagonal dominance)**.** Along the interior locus (6), the Schur complement of the renewable coordinate is negative,

$$\Psi'(K_R) = \Pi_{K_R K_R} - \begin{pmatrix} \Pi_{K_R K_F} & \Pi_{K_R K_M} \end{pmatrix} \mathcal{M}^{-1} \begin{pmatrix} \Pi_{K_R K_F} \\ \Pi_{K_R K_M} \end{pmatrix} < 0. \tag{8}$$

Condition 1 is a *single scalar* inequality—the second-order condition of the one-dimensional reduced problem $\Psi(K_R) = 0$, i.e. negative diagonal dominance of the Hessian in the renewable row after the (K_F, K_M)-block has been eliminated. It is strictly weaker than global concavity (which submodularity rules out) and it is exactly what lapses, becoming an equality, at the power-law knife-edge where $\Psi \equiv$ const.

[2] T1b: the own partial is strictly negative at $K_i = 60$ with the others moderate.

[3] T1c: $\det \mathcal{M}$ matches the closed form and $\operatorname{eig} \mathcal{M} < 0$ at the solved optimum, on Beta(2, 2) and Beta(2, 3) scale mixtures.

Theorem 1 (Existence, uniqueness, optimality)**.** *Under Assumption 1 and Condition 1, the profit* Π *has* at most one *interior critical point; it exists whenever the program's optimum is interior (the cutoffs* (5) *lie strictly in* $(0,1)$ *with* $K_M > 0$*—the Week-1 large-FCL adoption regime,* $C_M < \bar{C}_M$ *and* C_R *not so large that* $K_R^* = 0$*), in which case it is the* unique *interior maximizer* (K_R^*, K_F^*, K_M^*)*, the Hessian is negative definite there, and* $d^*(x) = \min\{(K_F - x)_+, K_M\}$ *is the unique optimal dispatch.*

Proof. Any interior critical point solves $\nabla\Pi = 0$, hence (cutoff Lemma) lies on the locus (6) and satisfies $\Psi(K_R) = 0$. By (7) and the chain rule, $\Psi'(K_R)$ equals the Schur complement (8) (the total derivative of g_R along the locus, after eliminating (K_F, K_M) through $g_M = g_F = 0$). Lemma 2 gives $\mathcal{M} \prec 0$, so $\mathcal{M}^{-1} \prec 0$ and the quadratic form in (8) is ≥ 0; Condition 1 makes $\Psi'(K_R) < 0$, so Ψ is strictly decreasing and has *at most one* zero—hence at most one interior critical point. When the optimum is interior (the stated regime), a maximizer exists by coercivity (Proposition 1) and lies in the interior, so it is a critical point and therefore *the* unique one K_R^*; (6) returns unique K_F^*, K_M^*. By the Schur determinant identity $\det\mathcal{H} = \det\mathcal{M} \cdot \Psi'(K_R)$ with $\det\mathcal{M} > 0$ and $\Psi' < 0$, the leading-principal-minor signs alternate, so $\mathcal{H} \prec 0$ at the optimum; the dispatch claim is Lemma 1.[4] □

Remark 1. The triangle is the whole point. Concavity would ask the full 3×3 Hessian to be negative definite as a joint hypothesis; the recursive structure instead *derives* negative definiteness from two cheap facts—an interiority-only 2×2 block (Lemma 2) and one scalar slope (Condition 1)—which is why the submodularity in (K_R, K_F) is no obstruction.

3. The shift-ratio condition, uniform along the homotopy

Week-1 Theorem 1 signs $K_R^* > K_R^\circ$ by integrating $dK_R^*/dC_M \leq 0$ downward from $\bar{C}_M$. The cofactor sign it integrates is

$$\mathrm{cof}_{K_M K_R}(\mathcal{H}) \propto f(K_F^*) f(a) \big[g(a) - g(K_F^*)\big], \qquad g(x; K_R) := \frac{F_{K_R}(x; K_R)}{f(x; K_R)}, \quad a = K_F^* - K_M^*,$$

so the step needs the *realized two-point endpoint inequality* $g(a) \geq g(K_F^*)$ to hold at *every* C_M on the path—even though the window $(a, K_F^*) = (K_F^* - K_M^*, K_F^*)$ moves with C_M. The following lemma supplies exactly that from a primitive, C_M-free hypothesis.

Condition 2 (Support-wide monotone shift ratio)**.** For each K_R in the equilibrium range, $x \mapsto g(x; K_R)$ is non-increasing on the support $(-K_R, \bar{D})$.

Lemma 3 (Uniform endpoint inequality along the homotopy)**.** *Suppose Condition 2 holds. Then for every* $C_M \in (0, \bar{C}_M)$*, at the interior optimum the window endpoints satisfy* $-K_R^* < a(C_M) < K_F^*(C_M) < \bar{D}$ *and*

$$g\big(a(C_M); K_R^*(C_M)\big) \;\geq\; g\big(K_F^*(C_M); K_R^*(C_M)\big), \tag{9}$$

with strict inequality wherever $g(\cdot; K_R^*)$ *is strictly decreasing. Hence* $\mathrm{cof}_{K_M K_R}(\mathcal{H}) \geq 0$ *and* $dK_R^*/dC_M \leq 0$ uniformly *on the path, and the Week-1 downward integration is justified by Condition 2 alone.*

Proof. At an interior optimum with $C_M < \bar{C}_M$ the adoption margin is strict, $K_M^* > 0$, so $a = K_F^* - K_M^* < K_F^*$. The cutoffs (5) give $F(a; K_R^*) = \varphi_1 \in (0,1)$ and $F(K_F^*; K_R^*) = \varphi_2 \in (0,1)$, so both endpoints lie strictly inside the support $(-K_R^*, \bar{D})$ (a point where $F \in (0,1)$ cannot be at an extended-CDF plateau). Condition 2 applied at $K_R = K_R^*$ to the two ordered interior points $a < K_F^*$ yields $g(a; K_R^*) \geq g(K_F^*; K_R^*)$, which is (9). This holds for every C_M because the hypothesis is on the whole support, independent of where the window sits; as C_M sweeps $(0, \bar{C}_M)$ the window

[4] T1c: $\Psi(K_R)$ is strictly decreasing on a scan of the locus with its unique zero at the solved K_R^* ($|\Psi(K_R^*)| < 10^{-6}$), and $\mathrm{eig}\,\mathcal{H} < 0$.

sweeps a sub-family of ordered interior pairs, all covered at once. The cofactor and derivative signs then follow from the Week-1 cofactor identity.[5] □

Remark 2. Condition 2 is the honest primitive behind Week-1 Condition CS, which was stated on the moving window "at each point of the equilibrium path." Lemma 3 shows the moving-window statement is *implied* by the static support-wide one, so the comparative static rests on a hypothesis one can check on F without first solving for the path. The power-law family has $g(x) = (\bar{D} - x)/(\bar{D} + K_R)$, affine decreasing for every α, hence satisfies Condition 2 trivially.

4. The emission condition for general F (keystone)

Week 1 closed the carbon sign only on the power-law/uniform knife-edge (a *corner* $K_R^* \in \{0, \infty\}$) and left a non-empty strict-fall window for general F *conjectural*. We close it at a *finite interior* optimum. Recall (Week 1, Prop. 4) the exact net fossil saving $\Delta := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ with

$$\mathbb{E}[\text{fossil}^*] = K_F^* - \tfrac{C_M}{R-c}(a - \mu^*_{\text{low}}), \qquad \mathbb{E}[\text{fossil}^\circ] = K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu^\circ_{\text{mid}}, \tag{10}$$

where $a = K_F^* - K_M^*$, $q^\circ = F(0; K_R^\circ)$, and $\mu^*_{\text{low}} = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^*, a)]$, $\mu^\circ_{\text{mid}} = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^\circ)]$. Carbon falls iff $\Delta > 0$, i.e. inequality (C) of Week 1 holds.

The endogenous condition. Substituting the equilibrium map $K_R^*(C_R), K_F^*(C_R), K_M^*(C_R)$ and $K_R^\circ(C_R), K_F^\circ(C_R)$ into (10) turns Δ into a function of primitives and C_R alone:

$$\Delta(C_R) = \Big[K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\mu^\circ_{\text{mid}}\Big] - \Big[K_F^* - \tfrac{C_M}{R-c}(a - \mu^*_{\text{low}})\Big], \tag{11}$$

all objects evaluated at the equilibrium. The difficulty is that the conditional means $\mu^*_{\text{low}}, \mu^\circ_{\text{mid}}$ and the capacities move with the shape of F.

Condition 3 (Two-zone density / tail-skew bound)**.** At the equilibrium, (a) $f(\cdot; K_R^*)$ is non-increasing on the FCL zone $(-K_R^*, a)$, and (b) $f(\cdot; K_R^\circ)$ is non-decreasing on the no-FCL zone $(0, K_F^\circ)$.

Part (a) is a left-tail/skew bound on *surplus* states (deep-surplus mass not too thin); part (b) a skew bound on the no-FCL mid zone. Both are zone-restricted monotone-density conditions, primitive given F.

Theorem 2 (General-F emission keystone)**.** *Suppose the program has a finite interior optimum (Theorem 1), Condition 2 holds (so $K_R^* > K_R^\circ$ by Lemma 3), and Condition 3 holds. Then*

$$\Delta \;\geq\; \Delta_{\text{LB}} := K_F^\circ\Big[(1-\varphi^\circ) + \tfrac{\varphi^\circ - q^\circ}{2}\Big] + \frac{C_M}{R-c}\cdot\frac{a + K_R^*}{2} - K_F^*, \tag{12}$$

and therefore carbon strictly falls ($\Delta > 0$) whenever $\Delta_{\text{LB}} > 0$. The bound is expressed in the equilibrium capacities and the distribution-free reliability φ° only. Caveat: *the condition $\Delta_{\text{LB}} > 0$ is a property of a capacity tuple, not a guarantee that any optimum attains it; Proposition 2 reports that it is essentially never attained at a solved interior optimum, so the value of* (12) *is its* form *(it pinpoints, via Lemma 4, the floor that defeats it), not its activation.*

Proof. Under Condition 3(a), a non-increasing density on $(-K_R^*, a)$ has conditional mean at most the interval midpoint, $\mu^*_{\text{low}} \leq (a - K_R^*)/2$ (the standard fact: a density non-increasing on $[\ell, u]$ has mean $\leq (\ell + u)/2$, with equality for the uniform); since the coefficient $-C_M/(R-c)$ on μ^*_{low} in (10) is negative, $\mathbb{E}[\text{fossil}^*] \leq K_F^* - \frac{C_M}{R-c}(a - \frac{a-K_R^*}{2}) = K_F^* - \frac{C_M}{R-c}\cdot\frac{a+K_R^*}{2}$. Under Condition 3(b), a non-decreasing density on $(0, K_F^\circ)$ has conditional mean at least the midpoint, $\mu^\circ_{\text{mid}} \geq K_F^\circ/2$, and since $\varphi^\circ - q^\circ > 0$, $\mathbb{E}[\text{fossil}^\circ] \geq K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)K_F^\circ/2$. Subtracting gives $\Delta \geq \Delta_{\text{LB}}$. The capacity channel $K_R^* > K_R^\circ$ (Lemma 3) is what makes the renewable term $\frac{C_M}{R-c}\cdot\frac{a+K_R^*}{2}$ large: a strictly

[5] T2: along a 12-point homotopy the realized gap $g(a) - g(K_F^*) \geq 0$ and $dK_R^*/dC_M \leq 0$ at every node; and $g(\cdot; K_R)$ is non-increasing on the full support for every sampled K_R.

larger K_R^* widens the FCL surplus zone and lowers $\mathbb{E}[\text{fossil}^*]$. Hence the listed hypotheses—endpoint shift-ratio (channel) plus the two-zone bound—are jointly sufficient for $\Delta > 0$.[6] □

Whether the sufficient bound is *activated* at a given optimum depends on the regime, decided by FCL capacity against renewable capacity. We can prove one half of the picture and must report the other half as a numerical finding. The provable half is a lower bound on FCL fossil.

Lemma 4 (Irreducible fossil floor)**.** *In the large-FCL regime ($K_M^* \geq K_R^*$), $\mathbb{E}[\textit{fossil}^*] \geq K_M^* - K_R^*$. Equivalently, the FCL cannot be fully powered by renewable surplus: the maximal surplus in any state is $(-\tilde{x})_+ \leq K_R^*$, so an FCL of size $K_M^* > K_R^*$ draws at least $K_M^* - K_R^*$ from fossil even in the deepest-surplus state.*

Proof. $\mathbb{E}[\text{fossil}^*] = \int_{-K_M^*}^{a}(1-F)\,dx \geq \int_{-K_M^*}^{-K_R^*}(1-F)\,dx = \int_{-K_M^*}^{-K_R^*} 1\,dx = K_M^* - K_R^*$, since $F \equiv 0$ below $-K_R^*$.[7] □

This floor is a genuine *additive* penalty, but it does not by itself sign $\Delta = \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$, because the capacity channel $K_R^* > K_R^\circ$ pushes the other way. Whether the floor (plus the tight-state utilization) outweighs the channel is, for general F, settled here only numerically.

Proposition 2 (Numerical resolution in the scanned family; conjecture in general)**.** *For the non-power-law Beta scale-mixture family $\tilde{x} = \tilde{D} - K_R\tilde{S}$ ($\tilde{D} \sim \bar{D}\,\text{Beta}(\alpha, \beta_D)$, $\tilde{S} \sim \text{Beta}(\alpha_S, \beta_S)$), an exhaustive numerical search over demand skews, capacity factors $\mathbb{E}[\tilde{S}]$ up to 0.8, and the $C_R \times C_M$ grid finds $\Delta \leq 0$ at* every large-FCL interior optimum located *(largest observed $\Delta^{\max} \approx -1.4$): the floor (Lemma 4) plus tight-state utilization numerically dominate the channel. We conjecture $\Delta < 0$ holds for all F in this family at large-FCL interior optima; the floor is the structural reason, but we do not have an analytic dominance proof. In the computed small-FCL examples ($K_M^* < K_R^*$, which the solver reaches only at small C_R) the floor vanishes yet $\Delta \leq 0$ for a different reason: at those small C_R the* no-FCL *grid is already near fossil-free ($K_F^\circ \to 0$), so little remains to save and the realized renewable build does not exceed K_R°; we flag this as a conjecture pending Role 4's small-FCL optimality analysis. Carbon strictly falls only at the cheap-renewable knife-edge* corner *$K_R^* \to \infty$ (power-law family, the window $(C_R^\circ, C_R^\dagger)$ of Week-1 Theorem 4(ii)), where $\mathbb{E}[\textit{fossil}^*] = 0 < \mathbb{E}[\textit{fossil}^\circ]$.*

Remark 3 (Status, and an earlier slip corrected). Week 1 conjectured carbon falls at an interior optimum but never exhibited one. Our contribution is to make the question sharp—Lemma 4 isolates the structural obstruction, Theorem 2 gives the (off-equilibrium) inequality whose violation is the obstruction, and an extensive search finds no interior optimum where carbon falls in the scanned family. We do *not* claim a universal non-existence theorem: the large-FCL statement is numerical (scan) plus the floor heuristic, and the small-FCL statement is a conjecture (its equilibrium structure is Role 4's, not derived here). The honest bottom line: interior carbon reduction is, at best, rare and unwitnessed; the only clean strict fall is the corner. (An earlier draft mis-stated the mechanism as "FCL draws fossil on surplus"; the fossil formula already credits surplus absorption—the correct obstruction is the floor of Lemma 4.)

5. Connectedness of the carbon-reduction set

Let $\mathcal{S} := \{C_R \geq 0 : \Delta(C_R) \geq 0\}$ be the set of renewable costs at which the program reduces carbon. Week 1 (Thm. 4) proved $\mathcal{S}$ bounded above for every F; it is non-empty via the cheap-renewable *corner* of the power-law family (Proposition 2: on $(C_R^\circ, C_R^\dagger)$, $K_R^* \to \infty$ gives $\mathbb{E}[\text{fossil}^*] = 0 < \mathbb{E}[\text{fossil}^\circ]$). On

[6] T3: the sufficient bound Δ_{LB} is computed at every interior optimum scanned; the algebra $\mu_{\text{low}}^* \leq (a - K_R^*)/2$, $\mu_{\text{mid}}^\circ \geq K_F^\circ/2$ under (a),(b) is checked directly.

[7] The fossil formula already credits surplus absorption: a state-by-state simulation that serves FCL from renewable surplus first matches $\int_{-K_M^*}^{a}(1-F)$ to ≈ 6 digits (T3, 6.18324 vs 6.18325). The floor is the *residual* fossil after that credit.

the Beta-mixture *interior*, by contrast, $\mathcal{S}$ may be empty (the same proposition). We now give the condition under which $\mathcal{S}$ is a single interval.

Condition 4 (Monotone renewable shadow value, MRSV)**.** At every C_R where the program is carbon-neutral, $\Delta(C_R) = 0$, the saving strictly down-crosses: $\Delta'(C_R) < 0$. Equivalently, at such a C_R the FCL fossil output rises strictly faster in C_R than the no-FCL output, $\frac{d}{dC_R}\mathbb{E}[\text{fossil}^*] > \frac{d}{dC_R}\mathbb{E}[\text{fossil}^\circ]$.

Economically: as renewables get dearer both programs shed them and both fossil outputs rise, but the FCL program—which holds strictly more renewables, $K_R^* > K_R^\circ$, and values them at the larger shadow weight $R > c$—sheds capacity of *higher* marginal fossil impact, so its fossil rises strictly faster. The shadow value $\sigma(K_R) = -d\,\mathbb{E}[\text{fossil}]/dK_R > 0$ being ordered, $\sigma^* > \sigma^\circ$, is exactly this strict down-crossing.

Theorem 3 (Single-interval carbon-reduction set)**.** (i) *$\mathcal{S}$ is bounded above: there is $B_0 < \infty$ with $C_R > B_0 \Rightarrow K_R^* = K_R^\circ = 0$ and $\Delta(C_R) = -U_0 < 0$, the strictly positive utilization penalty.* (ii) *$\mathcal{S} \neq \varnothing$ (the power-law corner, Proposition 2).* (iii) *Under Condition 4, Δ is single-crossing from above on $\{C_R : \Delta \geq 0\}$, so $\mathcal{S}$ is a single interval $[C_R^{\text{lo}}, C_R^\dagger]$ (possibly with $C_R^{\text{lo}} = 0$), where $C_R^\dagger = \sup \mathcal{S} < \infty$.*

Proof. (i) is Week-1 Theorem 4(i): above the finite renewable-benefit ceiling B_0 the renewable FOC has the corner $K_R^* = K_R^\circ = 0$, leaving only the distribution-free utilization effect, so $\Delta = -U_0 < 0$. (ii) is the corner clause of Proposition 2. For (iii), assume Δ is C^1 in C_R on the interior region (Theorem 1, IFT, under Condition 1); we show $\mathcal{S}$ is convex. Suppose not: then there are $C_R^{(1)} < C_R^{(2)} < C_R^{(3)}$ with $\Delta(C_R^{(1)}) \geq 0$, $\Delta(C_R^{(2)}) < 0$, $\Delta(C_R^{(3)}) \geq 0$. By the intermediate value theorem Δ has a zero in $(C_R^{(2)}, C_R^{(3)})$; let z be the smallest such zero. On $[C_R^{(2)}, z)$, $\Delta < 0$ (else there would be an earlier zero), and $\Delta(z) = 0$; hence Δ is non-decreasing as it reaches z from the left, $\Delta'(z) \geq 0$. This contradicts Condition 4, which forces the strict down-crossing $\Delta'(z) < 0$ at every zero z. So no interior dip exists and $\mathcal{S}$ is convex, i.e. an interval; boundedness (i) makes its right endpoint $C_R^\dagger = \sup \mathcal{S}$ finite, and strict down-crossing rules out any second crossing.[8] □

Remark 4. Condition 4 is close to its own conclusion: "strictly down-crosses at every zero" *is* single-crossing. The theorem's content is therefore the *packaging*—it identifies a primitive, checkable hypothesis (the shadow-value ordering $\sigma^* > \sigma^\circ$, the economic gloss above) that delivers single-crossing, given that Δ is C^1—rather than a derivation of single-crossing from nothing. Deriving $\sigma^* > \sigma^\circ$ from primitives, and C^1-ness of Δ across the regime boundary, are the parts we do not close here.

6. Sharpness: the necessary-and-sufficient channel condition and tight bounds

6.1. The sharp channel condition is a path integral.

Week 1 used pointwise-monotone g as a *sufficient* condition for $K_R^* > K_R^\circ$. The sharp condition is integral, not pointwise.

Theorem 4 (Necessary and sufficient channel condition)**.** *On the open large-FCL interior interval where Condition 1 holds ($\mathcal{H} \prec 0$), write $\lambda(C_M) := -dK_R^*/dC_M = \kappa(C_M)\, f(K_F^*) f(a)\, [\, g(a) - g(K_F^*)\,]$ with $\kappa(C_M) = (\beta + p - R)(R - c)/|\det \mathcal{H}| > 0$. For any compact subinterval $[C_M, C_M']$ of that open interval,*

$$K_R^*(C_M) - K_R^*(C_M') = \int_{C_M}^{C_M'} \lambda(t)\, dt, \tag{13}$$

and, letting $C_M' \uparrow \bar{C}_M$ where $K_R^*(C_M') \to K_R^\circ$ by continuity (Week 1), $K_R^*(C_M) > K_R^\circ$ if and only if $\int_{C_M}^{\bar{C}_M} \lambda > 0$. Pointwise $g(a) \geq g(K_F^*)$ (Condition 2) makes $\lambda \geq 0$ and is therefore sufficient*; it is* not necessary*, since the integral can be positive while the integrand is negative on part of the path.*

[8] T4: sweeping C_R on the witness family, $\Delta(C_R)$ has exactly one sign change ($+ \to -$), is non-increasing where $\Delta \geq 0$, and is negative at the largest C_R.

Proof. On the open interior interval Condition 1 gives $\mathcal{H} \prec 0$, so by the implicit function theorem $C_M \mapsto (K_R^*, K_F^*, K_M^*)$ is C^1 there, and $\lambda = -dK_R^*/dC_M$ is *continuous*. On a compact subinterval $[C_M, C_M']$ a continuous derivative is integrable, so K_R^* is C^1 (a fortiori absolutely continuous) and the fundamental theorem of calculus gives (13) exactly. The factorization of λ is the Week-1 cofactor identity divided by $|\det \mathcal{H}| > 0$. As $C_M' \uparrow \bar{C}_M$ the first FCL unit's marginal profit vanishes, $K_M^* \to 0$, and the system collapses to the no-FCL optimum, $K_R^*(C_M') \to K_R^\circ$ (Week 1, continuity at the adoption boundary). We *define* the improper integral $\int_{C_M}^{\bar{C}_M} \lambda$ as $\lim_{C_M' \uparrow \bar{C}_M} \int_{C_M}^{C_M'} \lambda$; by (13) this equals $\lim_{C_M' \uparrow \bar{C}_M} [K_R^*(C_M) - K_R^*(C_M')] = K_R^*(C_M) - K_R^\circ$, a finite limit—so the improper integral converges regardless of whether λ stays bounded as $C_M' \uparrow \bar{C}_M$ (where $\det \mathcal{H} \to 0$): convergence is inherited from the antiderivative's finite limit, not from integrability of λ. Positivity of $K_R^*(C_M) - K_R^\circ$ is then equivalent to positivity of $\int_{C_M}^{\bar{C}_M} \lambda$. Sufficiency of Condition 2 is $\lambda \geq 0$ pointwise; non-necessity is the elementary fact that an integral of a sign-changing integrand can be positive.[9] □

6.2. **The two-sided bounds are tight.**

Theorem 5 (Tight magnitude bounds)**.** *Fix the equilibrium capacities and q°, and range over all conditional laws consistent with the distribution-free cutoffs (these laws need not all be realized by an equilibrium F; the extremizers below are degenerate, not equilibrium laws). With Δ written as the affine map (10) of the conditional means $\mu_{\text{low}}^* \in (-K_R^*, a)$, $\mu_{\text{mid}}^\circ \in (0, K_F^\circ)$,*

$$\Delta_{\min} := K_F^\circ(1 - \varphi^\circ) - K_F^* \;\leq\; \Delta \;\leq\; K_F^\circ - K_F^* + \tfrac{C_M}{R-c}(a + K_R^*) := \Delta_{\max}, \tag{14}$$

and both bounds are the exact sup/inf of the affine map, attained in the limit: $\Delta_{\min}$ as the FCL conditional law concentrates at the upper endpoint ($\mu_{\text{low}}^* \uparrow a$) and the no-FCL law at the lower endpoint ($\mu_{\text{mid}}^\circ \downarrow 0$); $\Delta_{\max}$ as the laws concentrate oppositely ($\mu_{\text{low}}^* \downarrow -K_R^*$, $\mu_{\text{mid}}^\circ \uparrow K_F^\circ$, $q^\circ \downarrow 0$).

Proof. Δ is affine and monotone in each conditional mean: $\partial\Delta/\partial\mu_{\text{low}}^* = -C_M/(R-c) < 0$ and $\partial\Delta/\partial\mu_{\text{mid}}^\circ = \varphi^\circ - q^\circ > 0$. A conditional mean ranges over the open interval between its endpoints, with the endpoints approached by laws concentrating there (e.g. a two-point law placing mass $\to 1$ at the endpoint while preserving the cutoff probability). Plugging the extreme means into (10) gives (14), and the attaining sequences are the concentrating laws just named; the bounds are therefore the exact sup/inf, i.e. tight.[10] □

7. Numerical validation

All identities are checked in `verify_week2.py` against adaptive quadrature of the profit integral (1), on non-power-law Beta scale mixtures $\tilde{x} = \tilde{D} - K_R\tilde{S}$. T1 confirms the recursive triangle (interiority-only (K_F, K_M)-block, scalar Schur, $\mathcal{H} \prec 0$, unique Ψ zero); T2 the uniform endpoint inequality along the homotopy; T3 the exact endogenous identity, the sufficient-bound algebra, and the Proposition 2 scan ($\Delta < 0$ at every large-FCL interior optimum, $\Delta^{\max} \approx -1.4$); T4 the corner non-emptiness $C_R^\circ < C_R^\dagger$, the large-cost anchor, and the single interval; T5 the FTC/path-integral identity and the tight bounds $\Delta_{\min} \leq \Delta \leq \Delta_{\max}$ attained at the endpoints. The suite reports a single pass count.

[9] T5: on the interior band $\int_{C_M}^{C_M'} (-dK_R^*/dC_M)\, dt$ matches $K_R^*(C_M) - K_R^*(C_M')$ to $\sim 2\%$ (4.21 vs 4.13); the sufficient-not-necessary logic is shown by a sign-mixed integrand with positive integral. A *realized* F with $\lambda < 0$ on a sub-path is Role 3's reversed-channel counterexample, outside this deliverable.

[10] T5b: $\Delta(\mu_{\text{low}}^* = a, \mu_{\text{mid}}^\circ = 0) = \Delta_{\min}$ and $\Delta(-K_R^*, K_F^\circ) = \Delta_{\max}$ exactly, and the realized Δ lies strictly inside.